REGIS: Refining Generated Videos via Iterative Stylistic Redesigning

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Research Article

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Refining Generated Videos via Iterative Stylistic Redesigning

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Abstract

In recent years, generative models have made impressive advancements towards realistic output; in particular, models working in the modalities of text and audio have reached a level of quality at which generated text or audio cannot be easily distinguished from real text or audio Despite these revolutionary advancements, the synthesis of realistic and temporally consistent videos is still in its inception. In this paper, we introduce a novel approach to the creation of realistic videos that focuses on improving the generated video in the latter steps of a video generation process. Specifically, we propose a framework for the iterative refinement of generated videos through repeated passes through a neural network trained to model the spatio-temporal dependencies found in real videos. Through our experiments, we demonstrate that our proposed approach significantly improves upon the generations of text-to-video models and achieves state-of-the-art results of the UCF-101 benchmark; removing the spatio-temporal artifacts and noise that make synthetic videos distinguishable from real videos. In addition, we discuss the ways in which one might augment this framework to achieve better performance.

Keywords: video generation, video processing, video inpainting, generative models, neural networks

1 Introduction

Generative models have made impressive advancements in the past few years. Specifically, models focused on the generation of images from text have made groundbreaking advancements [1–3], with these models being capable of generating photo-realistic, aesthetically pleasing images.

In attempts to replicate these achievements in the domain of videos, multiple approaches have been proposed. Saito et al. [4] propose an extension to the traditional paradigm of the Generative Adversarial Network (GAN) [5] that allows one to operate temporally. Ho et al. [6] propose the extension of the u-net diffusion architecture typically used for image generation to image domain through the addition of temporal attention. Luo et al. [7] propose the decomposition of video diffusion into base frames and residual frames, allowing for the similarities between video frames to be captured more effectively. These methods, while having demonstrated an ability to generate conceptually correct vidoes, still struggle with accurately replicating the spatio-temporal relationships that are found in real videos.

To allow for the more effective creation of realistic videos, we propose a paradigm that focuses on the capturing and improving the initial generated video in the latter parts of the video generation process. We intuit that an image-to-image (or video-to-video) model trained on a task that requires a fine-grained understanding of space-time may be able to capture spatio-temporal relationships more effectively than video generators, due to not having the burden of creativity. Given a reasonably capable model, performing repeated passes with the aforementioned model may lead to the removal of flaws in synthetic videos; more specifically, traits that allow for a clear distinction between synthetic videos and real videos, such as the constant minor pixel changes that are largely present in generated videos and unexpected color changes.

With this reasoning, in combination with the success of Autoencoders and inpainting techniques [8, 9], we modify the standard video generation process to include a module that uses repeated passes to redesign the style of generated videos, which we call REGIS. In combination with the models created by previous works, we show that REGIS facilitates state-of-the-art performance on video generation.

Our contributions can be summarized as follows:

1. We propose a novel framework for the iterative refinement of videos created by text-to-video models.
2. We perform extensive tests on the efficacy of the proposed framework, using several different models, showing that the the framework achieves state-of-the-art (SOTA) results on the UCF101 video generation benchmark.
3. We propose several directions for future research, allowing for further improvement upon the proposed paradigm.

2 Related work

2.1 Video Generation

Assisted by the rapid progress of text-to-image models, text-to-video models have made considerable improvements in both quality and efficiency. Through the years, several frameworks for the generation of videos from text have been proposed, making use of both Generative Adversarial Networks (GANs)[4, 10] and Diffusion technology [6, 7, 11].

2.1.1 GANs

Since the field's inception, Generative Adversarial Networks (GANs)[5] have been a cornerstone in the field of video generation. VGAN [10], TGAN [4], and MoCoGAN [12], demonstrate the various approaches that have been used in pursuit of photorealistic video generation, with VGAN proposing a dual-stream architecture for the modeling of static and moving foreground videos, TGAN proposing a two-step generation process, involving the consecutive generation of temporal then spatial representations, and MoCoGAN introducing a motion-content disentanglement framework.

2.1.2 Diffusion Models

Recently, due to the success of text-to-image models, state-of-the-art (SOTA) text-to-video models have increasingly explored the paradigm of diffusion: a technique in which stochastic processes are used to model the dynamics of the progressive noising of a given input.

A general equation to represent the diffusion process can be given as:

$$x_t = \sqrt{1 - \beta_t} \cdot x_{t-1} + \sqrt{\beta_t} \cdot \epsilon_t \tag{1}$$

Where:

- x_t is the state of the data at time step t.
- x_{t-1} is the state of the data at the previous time step.
- β_t is a time-dependent noise scale factor.
- ϵ_t is a noise term sampled from a Gaussian distribution at time step t.

The neural network typically predicts the noise scale factor β_t and guides the transformation of x_{t-1} to x_t over multiple steps.

Ho et al. [6] explore a natural extension of the image diffusion paradigm through the use of factored spatio-temporal attention, allowing for joint image-video training. Ge et al. [11] explore the use of carefully designed noise in making use of the innate similarities between the domains of images and videos. Luo et al. [7] decompose the diffusion paradigm, using 2 generators - a base and residual generator - instead of 1, to better account for content redundancy and temporal correlation in the diffusion process.

2.2 Inpainting

Inpainting, the task of filling in the missing portions of a images or videos with plausible content, has been a subject of extensive research in both image and video processing domains.

2.2.1 Image Inpainting

From early inpainting techniques, which replicated known textures in the missing part of a given image, one example of which being PatchMatch [13], image impainting techniques have evolved to use deep-learning based approaches. Recent works have evaluated the efficacy of generative adversarial networks [5] as well as end-to-end architectures for the task of image inpainting. Yu et al. [14] propose the use of contextual attention for inpainting by propagating relevant features from the uncorrupted regions of the image to the missing portions.

2.2.2 Video Inpainting

Being the natural extension of image inpainting, the task of video inpainting has also seen rapid advancements in recent years. Wang et al. [15] propose a network architecture that makes use of both 2D and 3D convolutions for the creation of content for corrupted regions within videos. Xu et al. [16] use extracted optical flows to guide the filling of holes in a target video. Kim et al. [17] make use of a recurrent neural network to aggregate temporal features accross video frames. Zeng et al. [18] introduce the use of a joint spatio-temporal transformer Network (STTN) for video inpainting. Liu et al. [9] develop several novel operations, including 'soft-split' and 'soft-composition' to fix the issue of blurry inpainted features, using them in a transformer model that allows for sub-patch level information interaction.

2.3 Autoencoders

Autoencoders, models that aim to represent a high-dimensional piece of data as a lower-dimensional encoding, have been used in various applications as of recent [19]. Whilst the basic structure of an autoencoder, composed of an encoder and a decoder, has not undergone much change in recent years, various modifications have been made since its creation. In denoising autoencoders [20], the input is disturbed by some noise, and the autoencoder is trained to predict the de-noised version of the input. Variational Autoencoders (VAEs) [21] introduce a probabilistic graphical model with an associated variational inference procedure, enabling the generation of new data points. Vector-Quantized Variational Autoencoders (VQ-VAEs) [22] utilize a discrete latent representation through vector quantization, enabling more structured representations.

3 Methodology

3.1 The Standard processes

As displayed in Figure 1, to generate videos, most previous works [4, 6, 7, 10, 11] have adopted one of three methods:

1. Directly generating videos using a singular model.
2. Generating key-frames with a base text-to-video generator followed by a cascade of spatial and temporal upscalars.
3. Using a base text-to-video generator that operates in latent space, which is then remapped into a high-dimensional video-space.

We propose an addition to these processes in which the generated video is repeatedly passed through a given model, allowing for the iterative refinement of the videos

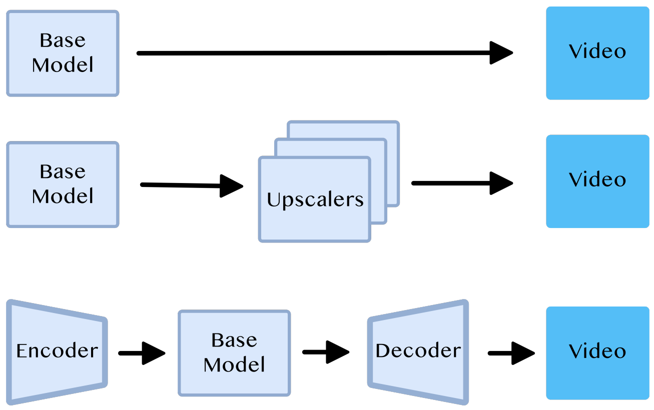

Fig. 1 Visualization of the standard processes used to generate high-resolution videos from text.

generated by a base generator. To evaluate this paradigm, we test 3 models: A VQ-VAE, a U-net inpainter that fills in partially-obscuring masks (created using gaussian blur), and FuseFormer [9], a video inpainter that fills in fully-obscuring masks. We denote the VQ-VAE, U-net, and FuseFormer models as **REGIS-VQ**, **REGIS-U** and **REGIS-Fuse** respectively.

3.2 Repeated Inpainting

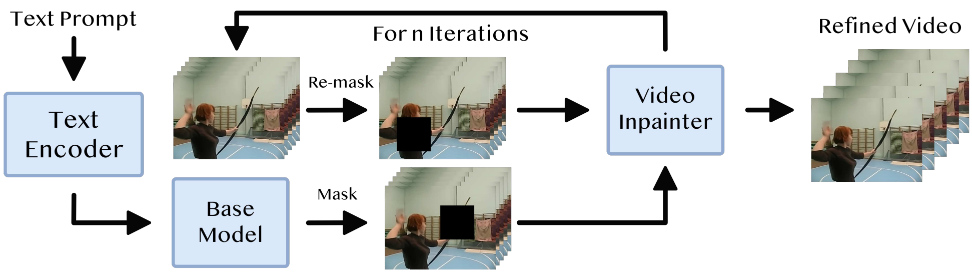

Fig. 2 Visualization of REGIS with inpainting. Given an encoded text prompt, a base text-to-video model, such as VideoFusion [7] will convert the text to an initial video. That initial video is then masked and inpainted for n iterations, resulting in a refined video.

For the process of repeated inpainting, given a batch of videos and a chosen masking type, each video in the batch is masked and subsequently passed though an inpainter to fill in the masked sections (note: we do not insert the previously masked portion back into the original video, as we find that it results in lower performance). We repeat this process for I iterations.

Algorithm 1 Repeated Inpainting for Video Refinement

```
procedure REGIS(Video, Inpainter, Iterations, Mask)
    for Iteration in Iterations do
        Masked video, mask ← Mask(Video)
        Inpainted video ← Inpainter(Masked video)
    end for
    return Video
end procedure
```

3.3 Repeated Quantization

For the process of REGIS-VQ, which we call repeated quantization, a batch of videos is passed through a VQ-VAE repeatedly for I iterations.

Algorithm 2 Repeated Quantization for Video Refinement

```
procedure REGIS(Video, VQ-VAE, Iterations,)
    for Iteration in Iterations do
        Video ← VQ-VAE(Video)
    end for
    return Video
end procedure
```

3.4 Architectures

3.4.1 REGIS-U

Extending the 2D U-net architecture commonly used for image diffusion, Ho et al. [6] make use of factored space-time attention by using space-only 3D convolutions (3D convolutions in which the spatial axis of the kernel is 1) with spatial attention followed by temporal attention. To make this model more suitable for inpainting, we add 3D attention mechanisms to the end of each downscaling block and employ a dual channel approach in the upscaling portion of the architecture. In each upscaling block, we send the initial data separately through 2 modules, then combine the 2 sets of information with a channel combiner module; the first, a module composed of a resnet block and factored spatio-temporal attention, and the second: a module composed of a resnet block and 3D attention. The makeup of our channel-combiner block can be seen in Figure 3.

3.4.2 REGIS-VQ

For our VQ-VAE, we use the same architecture elucidated by Yan et al. [23], with the encoder being structured with a sequence of 3D convolutions and subsequent attention residual blocks, each of which adopting a configuration that makes use of LayerNorm and axial attention layers. Conversely, the decoder's architecture mirrors

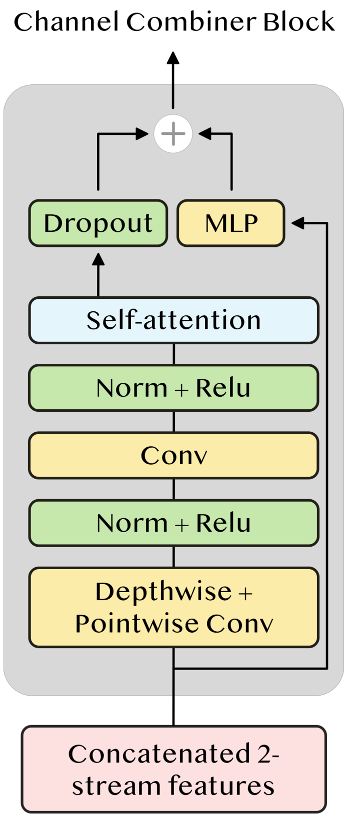

Fig. 3 Visualization of the Channel-Combiner block.

the encoder, starting with attention residual blocks and making use of a series of 3D transposed convolutions for spatial-temporal upsampling. Additionally, for all axial attention layers, learned spatio-temporal embeddings serve as position encodings.

3.4.3 REGIS-Fuse

For REGIS-Fuse, we use the exact architecture used in [9].

4 Experiments

4.1 Dataset

To evaluate the proposed paradigm, we perform experiments using the UCF101 dataset, a categorical dataset that, while originally being designed for the task of action recognition, has become a widely adopted benchmark for the evaluation of text-to-video models. The UCF101 dataset is composed of 13,320 videos each belonging to one of 101 categories, each representing a different action.

4.2 Metrics

For quantitative evaluation against other frameworks, we follow previous works [7, 11], using Frechet Video Distance (FVD)[24] as our main metric.

4.3 Training Details

For the training of our augmented version of the U-net model proposed by Ho et al [6], we train a U-net model with the originally proposed architecture and use it to initialize our augmented model. Both the model used for initialization and final model are trained using the UCF101 dataset.

For FuseFormer [9], we make use of the publicly available pre-trained weights.

For our VQ-VAE, we follow the same training regimen specified in [23], alongside pre-training on the Kinetics-400 dataset [25].

4.4 Evaluation Details

For our quantitative comparison against other approaches, we follow the evaluation procedure outlined in [26] with the final evaluation being done with 2048 clips of size 16x128x128. In addition, we set iterations to 2, 2, and 8 for REGIS-VQ, REGIS-U, and REGIS-Fuse respectively, alongside mask sizes of 4x65x65x3 and 4x16x16x3 with four masks per iteration for REGIS-Fuse and REGIS-U respectively.

4.5 Results

Table 1 Quantitative analysis of the proposed framework using VideoFusion [7] as the base text-to-video model. ↓ denotes the higher the better. I denotes the number of iterations.

Method	Resolution	FVD↓
VideoFusion[7]	16 x 64 x 64	130.19 ± 6
REGIS-VQ$^{I=2}$	16 x 64 x 64	129.39 ± 6
REGIS-U$^{I=2}$	16 x 64 x 64	115.36 ± 6
REGIS-Fuse$^{I=8}$	16 x 64 x 64	125.32 ± 6

In comparing the fidelity of videos before and after the use of our proposed paradigm we find that REGIS-U and REGIS-Fuse lead to measurable improvements in Frechet Video Distance (FVD). Specifically, REGIS-U$^{I=2}$ achieves the best performance, reducing the FVD to 115.36 ± 6, a noticeable improvement over the baseline VideoFusion model, which has an FVD of 130.19 ± 6. REGIS-Fuse$^{I=8}$ also shows better performance than the baseline with an FVD of 125.32 ± 6.

Upon closer inspection of the videos produced by REGIS-VQ, we find that for roughly 55% of the UCF101 classes, the resultant FVD is reduced. More specifically, as depicted in Table 2, we find that REGIS-VQ leads to significant improvements in classes with large amounts of movement, including athletic events such as high-jump and pole vault, and significant deterioration in classes with low amounts of movement, such as knitting and playing the cello. Notably, REGIS-VQ achieves an FVD 179.3% better than VideoFusion's [7] original FVD on the pole vault class, underscoring its ability to drastically improve video fidelity on high-movement videos. In addition, computing FVD using two iterations with REGIS-VQ for high-movement classes and 0 iterations for low-movement classes results in a value of 108.19, a 17% decrease from VideoFusion's initial FVD.

Table 2 Comparison of REGIS-VQ FVD between high-movement and low-movement classes with VideoFusion [7] as the base text-to-video model. Note a lower FVD represents a higher-quality video.

Class	VideoFusion	REGIS-VQ
High-movement classes		
Pole Vault	370.06	76.684
High Jump	259.67	114.53
Hammer Throw	193.21	120.92
Lunges	146.91	92.63
Low-movement classes		
Playing Tabla	112.19	160.82
Playing Guitar	86.93	174.44
Knitting	37.106	127.08
Playing Cello	87.58	131.81

In addition, we quantitatively compare REGIS against several competitive methods including VideoFusion[7], PYoCo[11], and VDM[6] on class-conditioned video generation using their reported FVD scores. Table 3 shows that, using VideoFusion as the base text-to-video model, our method outperforms all previous methods on the UCF-101 dataset.

Table 3 Quantitative analysis of REGIS compared to recent methods on the UCF101 dataset. ↑ denotes the higher the better and ↓ the lower the better

Method	Resolution	FVD↓
TGANv2 [27]	16 x 128 x 128	1209
TATS [26]	16 x 128 x 128	332
VideoFusion [7]	16 x 128 x 128	173
REGIS-Fuse	16 x 128 x 128	141

Due to qualitative observations, that REGIS-VQ, improves videos through the reduction of spatio-temporal noise, which according to Zhao et al. is one of the main factors that impact perceived video quality, we conduct an additional experiment using measured mean squared error (MSE) to analyze the degree to which REGIS removes spatio-temporal noise. In this experiment, we use MSE between consecutive frames of 100 16x64x64 still videos generated by VideoFusion [7] before and after being passed through REGIS-VQ to measure the amount of spatio-temporal noise. For the generation of these videos, we use prompts aimed at the generation of videos that do not contain movement such as "A still red apple on a white background"; in addition, we manually check all 100 videos to make sure no actions occur in the generated videos, to ensure that all pixel movement is spatio-temporal noise. Table 4 shows that REGIS-VQ decreases unprecedented movement in still videos, thereby, significantly increasing the quality of generated videos.

Table 4 Quantitative analysis of REGIS using still videos. ↓ denotes the lower the better

Method	Resolution	MSE↓
VideoFusion [7]	16 x 64 x 64	4991.59
REGIS-VQ	16 x 64 x 64	3993.24

4.6 Limitations

1. **Iterations:** Due to hardware limitations we are unable to fully investigate the scalability of the proposed framework, testing only 2 iterations for REGIS-VQ and REGIS-U and up to 8 iterations for REGIS-Fuse. We do note that, experimentally, we find that the resultant FVD of REGIS-Fuse scales positively with iterations between $Iterations = 1$ and $Iterations = 8$.
2. **Dataset Limitations**: All tests, barring the experiment measuring REGIS-VQ noise reduction, were conducted using the UCF101 dataset which, despite its robustness, is relatively small when compared to all possible video actions.

5 Conclusion

We propose a novel module to be added into the standard video generation process that refines a video generated by an initial text-to-video model with repeated passes. We demonstrate that the proposed addition allows for significant improvement for the task of text-to-video generation, reducing Frechet Video Distance (FVD)[24] on UCF101 by 17%, reducing spatio-temporal noise by 20%, and achieving state-of-the-art performance in terms of FVD on the UCF-101 dataset.

6 Future Work

For future research, we present two promising directions:

6.1 Other Models

In our experiments, we show that our proposed paradigm is capable of achieving state of the art results. In the future, we aim to evaluate the overall performance and scalability of REGIS when used with other models, such as denoising autoencoders and image inpainters.

6.2 Incorperation of Generative Adversarial Networks

In the future, the use of a Geneative Adversarial Network (GAN) [5] system, in which an inpainter would serve as the generator, would allow for a loop that would work in such a way that the input video would be refined until it became a photo-realistic video.

Supplementary information. All supplementary code and data associated with this research is publicly accessible and can be found at this GitHub link. The code is distributed under the MIT License. For any additional data used in this research that are not publicly accessible, such as model weights, please contact the corresponding author for access.

Declarations

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- Authors' contributions: Not applicable.

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